

- 1 (a) Fig. 1.1 is a diagram of part of a Bowman's capsule and a glomerular capillary.

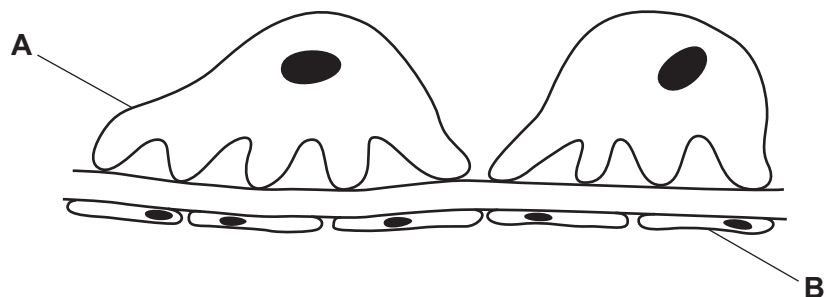


Fig. 1.1

- (i) Identify structures **A** and **B**.

A

B [2]

- (ii) The glomerular filtrate is produced in the Bowman's capsule by the process of ultrafiltration.

State the conditions required for ultrafiltration.

.....
.....
.....
.....
..... [2]

- (b) As the newly formed glomerular filtrate passes along the proximal convoluted tubule, selective reabsorption takes place. The fluid remaining in the proximal convoluted tubule eventually forms urine.

Table 1.1 lists three of the components of blood plasma that enter the glomerulus in a healthy person.

Complete Table 1.1 by writing:

- **increased**, if the component is present and is higher in concentration than in blood plasma
- **decreased**, if the component is present and is lower in concentration than in blood plasma
- **same**, if the component is present and is of the same concentration as blood plasma
- **not present**, if the component is absent.

You may use each response once, more than once or not at all.

Table 1.1

component of blood plasma entering glomerulus	component in newly formed glomerular filtrate	component in urine
glucose		
large plasma proteins		
urea		

[3]

- (c) Sweating during exercise can lead to a response in the posterior pituitary gland and in the kidney.

State the response to sweating that occurs in the posterior pituitary gland **and** the response that occurs in the kidney.

.....

.....

.....

.....

..... [2]

[Total: 9]